Commercial cellular coverage test: Example: carrier cellular DAS, warehouse (completed example)

Example distribution warehouse, Arlington VA area (fictional) · Example County Building Official (example) · Owner spec: carrier coverage, Common preset · prepared 2026-09-30

Result: PASS

AT&T, below 3.3 GHz: **pass** · T-Mobile, below 3.3 GHz: **pass** · Verizon, below 3.3 GHz: **pass**

Pass criteria: Common: each required carrier's best cell RSRP ≥ -95 dBm (≥ -100 dBm from 3.3 GHz) with SINR ≥ 5 dB on 95% of the tested squares on each floor; carriers required: AT&T, Verizon, T-Mobile. Owner specs and RFPs cluster at RSRP –95 dBm over 95% for LTE and –100 to –105 dBm for C-band; SINR runs 5–10 dB in US documents. Guidance, not a carrier spec. Only readings taken with an instrument calibrated within 12 months of the reading are scored.

Building and people

Building	Example: carrier cellular DAS, warehouse (completed example) · Example distribution warehouse, Arlington VA area (fictional) · 1 floors · 120,000 sq ft · Traditional build, worst 5% (28 dB)
AHJ / fire marshal	Alicia Grant · Example County Building Official (example) · 703-555-0146 · agrant@buildingofficial.example
Building owner	Ray Kim · Sample Logistics LLC (example) · 703-555-0151 · rkim@samplelogistics.example
Our technician	Jordan Lee · Sample RF Services (example) · 703-555-0109 · jlee@samplerf.example

Tester and instruments

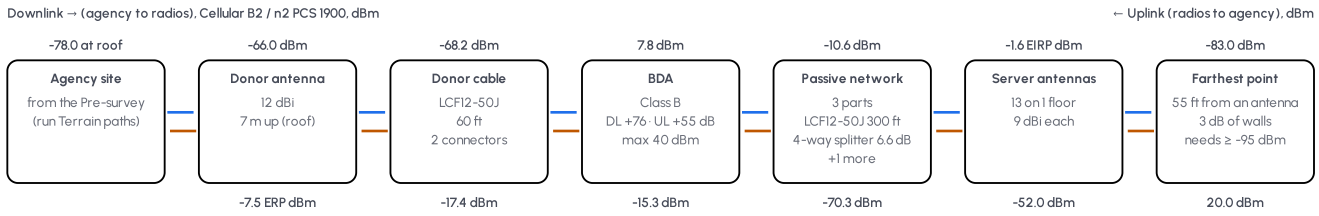
Tester	Jordan Lee (example)
FCC GROL	PG-EX-0001 (example)
NICET	Level III (example)
Test dates	2026-09-22 to 2026-09-23

Instrument	Serial	Calibrated	Certificate	Status
Nextivity COMPASS XR	EX-2001	2026-05-01	Calibration certificate - Nextivity COMPASS XR EX-2001 (example).pdf, to 2027-05-01	current

Compliance

AHJ permit	2026-07-01
Design approved	2026-07-24
Acceptance test passed	2026-09-30
Battery test	2026-09-23
Annual retest due	2027-09-30
Compared with	the acceptance test of 2026-09-30

System diagram: signal path, start to finish



Levels are the Cellular B2 / n2 PCS 1900 link budget's worst case (1990 MHz down, 1910 MHz up); every band is in the table below. Blue: what reaches the radios inside; orange: what reaches the agency's receiver.

Band	Systems	At roof dBm	BDA out dBm	Antenna EIRP dBm	Farthest point dBm	Needs dBm	Uplink at site dBm	Checks
Cellular B2 / n2 PCS 1900	Verizon, AT&T, T-Mobile	-78.0	7.8	-1.6	-83.0	-95.0	–	pass, 1 warn
Cellular B13 / n13 Upper 700	Verizon	-88.0	-3.3	-8.8	-81.7	-95.0	–	pass, 1 warn
Cellular B71 / n71 600 MHz	T-Mobile	-84.0	-2.3	-7.2	-78.9	-95.0	–	pass, 1 warn

Rooftop donor	RSRP dBm	SINR dB	Band	Azimuth	Height ft	Measured
AT&T	-78.0	11.0	2	215°	38	2026-06-16
Verizon	-88.0	6.0	13	160°	38	2026-06-16
T-Mobile	-84.0	7.0	71	20°	38	2026-06-16

Antennas by floor: Floor 1: 13

Commercial cellular coverage

Target: Common: RSRP ≥ -95 dBm (≥ -100 dBm from 3.3 GHz), SINR ≥ 5 dB on 95% of the tested squares, carriers: AT&T, Verizon, T-Mobile. Owner specs and RFPs cluster at RSRP –95 dBm over 95% for LTE and –100 to –105 dBm for C-band; SINR runs 5–10 dB in US documents. Guidance, not a carrier spec. Each square counts the carrier's best cell.

Floor	Carrier	Layer	Tested	Pass	%	Result
1	AT&T	below 3.3 GHz	20	20	100%	pass

Floor	Carrier	Layer	Tested	Pass	%	Result
1	T-Mobile	below 3.3 GHz	20	20	100%	pass
1	Verizon	below 3.3 GHz	20	20	100%	pass

Rooftop donor

Carrier	RSRP dBm	SINR dB	Band	Azimuth	Height ft	Measured	Against Providence §3.9
AT&T	-78	11	2	215°	38	2026-06-16	meets −85 dBm target
Verizon	-88	6	13	160°	38	2026-06-16	minimum only (−95 dBm)
T-Mobile	-84	7	71	20°	38	2026-06-16	meets −85 dBm target

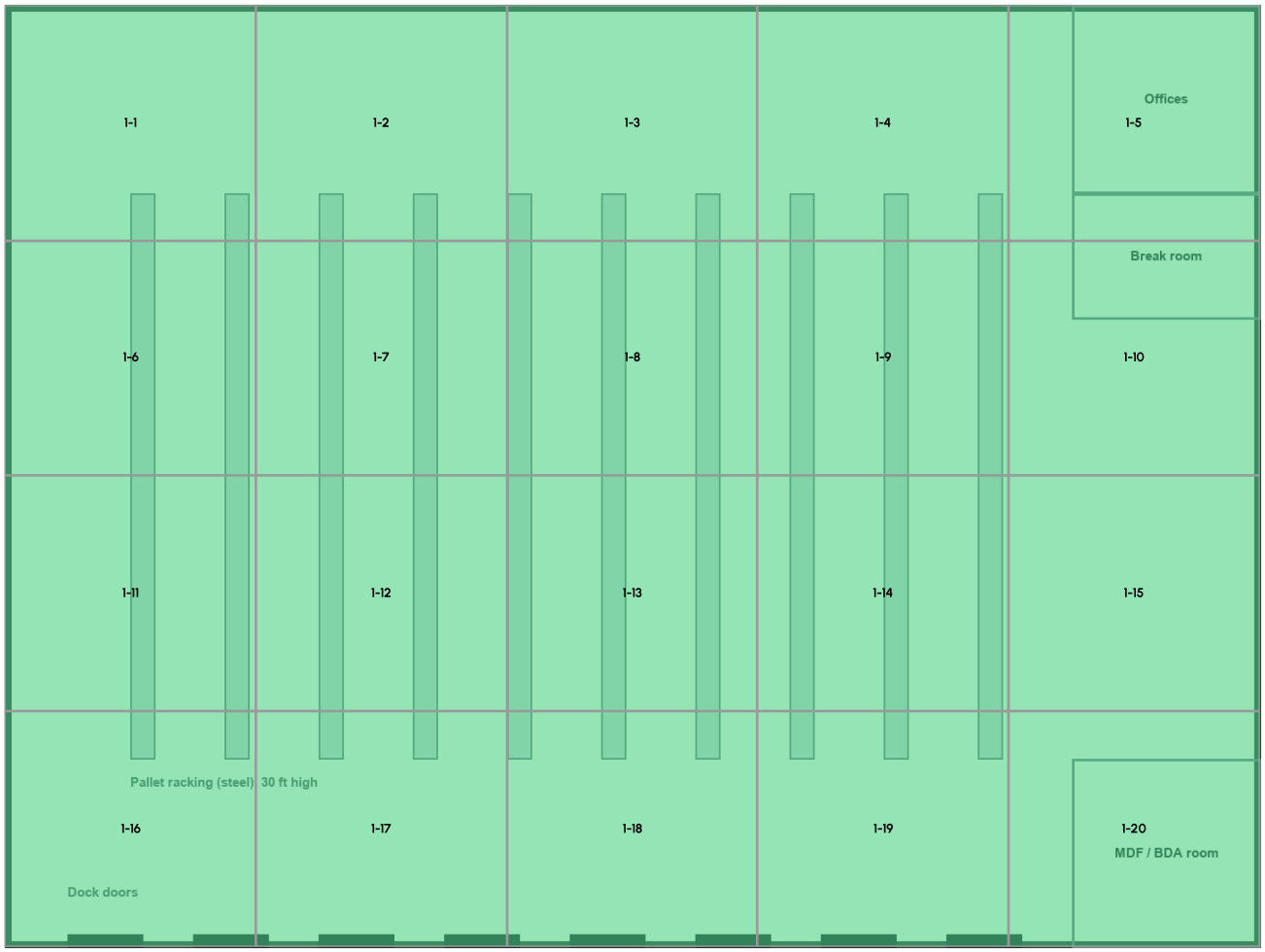
Carrier approvals

Carrier	Path	Asked	Consent	Contact / reference
AT&T	Industrial booster or off-air DAS	2026-06-22	2026-07-20	AT&T DAS team (example), ref ATT-EX-0442: B2 consented
Verizon	Industrial booster or off-air DAS	2026-06-22	2026-07-28	Verizon account manager (example): B13 consented
T-Mobile	Industrial booster or off-air DAS	2026-06-23	2026-08-11	Rebroadcast Agreement signed (example), ref TMO-EX-1187: B71

Floor 1: cellular coverage

Cellular (Common): PASS · AT&T, below 3.3 GHz: 20/20 (100%) pass · T-Mobile, below 3.3 GHz: 20/20 (100%) pass · Verizon, below 3.3 GHz: 20/20 (100%) pass · target RSRP ≥ -95 dBm, SINR ≥ 5 dB on 95%

Cellular: every required carrier (AT&T, below 3.3 GHz; T-Mobile, below 3.3 GHz; Verizon, below 3.3 GHz)



Example distribution warehouse - Floor 1 (fictional plan for the RF Recon DAS example)
400 ft x 300 ft, 32 ft clear height. Not a real building. Metal roof deck and tilt-up concrete walls.

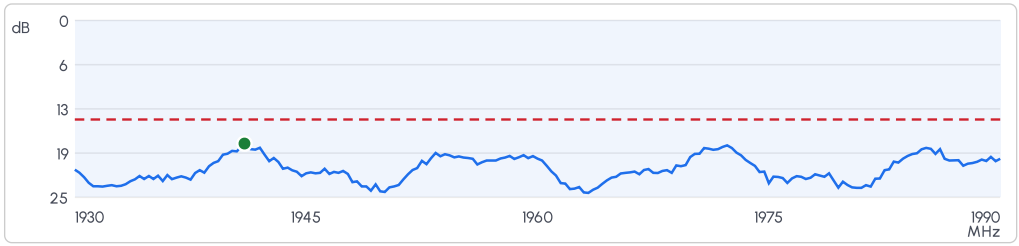
Square	AT&T, below 3.3 GHz	T-Mobile, below 3.3 GHz	Verizon, below 3.3 GHz	Cellular
1-1	-66 dBm · SINR 24.7 · PCI 212 · B2 ✓	-63 dBm · SINR 26.3 · PCI 301 · B71 ✓	-68 dBm · SINR 21.8 · PCI 88 · B13 ✓	pass
1-2	-77 dBm · SINR 18.6 · PCI 212 · B2 ✓	-71 dBm · SINR 19.7 · PCI 301 · B71 ✓	-75 dBm · SINR 18.1 · PCI 88 · B13 ✓	pass
1-3	-85 dBm · SINR 13.5 · PCI 212 · B2 ✓	-77 dBm · SINR 17.3 · PCI 301 · B71 ✓	-78 dBm · SINR 15.9 · PCI 88 · B13 ✓	pass
1-4	-75 dBm · SINR 17.4 · PCI 212 · B2 ✓	-72 dBm · SINR 19.9 · PCI 301 · B71 ✓	-75 dBm · SINR 19.1 · PCI 88 · B13 ✓	pass
1-5	-66 dBm · SINR 24.5 · PCI 212 · B2 ✓	-65 dBm · SINR 24.5 · PCI 301 · B71 ✓	-69 dBm · SINR 22.1 · PCI 88 · B13 ✓	pass
1-6	-81 dBm · SINR 15.9 · PCI 212 · B2 ✓	-72 dBm · SINR 18.6 · PCI 301 · B71 ✓	-78 dBm · SINR 14.3 · PCI 88 · B13 ✓	pass
1-7	-79 dBm · SINR 17.4 · PCI 212 · B2 ✓	-73 dBm · SINR 18.7 · PCI 301 · B71 ✓	-77 dBm · SINR 17.7 · PCI 88 · B13 ✓	pass
1-8	-86 dBm · SINR 12.7 · PCI 212 · B2 ✓	-81 dBm · SINR 15.1 · PCI 301 · B71 ✓	-83 dBm · SINR 16.2 · PCI 88 · B13 ✓	pass
1-9	-81 dBm · SINR 16.9 · PCI 212 · B2 ✓	-79 dBm · SINR 16.3 · PCI 301 · B71 ✓	-76 dBm · SINR 18 · PCI 88 · B13 ✓	pass
1-10	-73 dBm · SINR 19.8 · PCI 212 · B2 ✓	-70 dBm · SINR 18.8 · PCI 301 · B71 ✓	-77 dBm · SINR 22 · PCI 88 · B13 ✓	pass
1-11	-79 dBm · SINR 17.9 · PCI 212 · B2 ✓	-74 dBm · SINR 20.2 · PCI 301 · B71 ✓	-77 dBm · SINR 19.3 · PCI 88 · B13 ✓	pass
1-12	-78 dBm · SINR 19.2 · PCI 212 · B2 ✓	-79 dBm · SINR 17.8 · PCI 301 · B71 ✓	-82 dBm · SINR 14.6 · PCI 88 · B13 ✓	pass
1-13	-84 dBm · SINR 16.5 · PCI 212 · B2 ✓	-80 dBm · SINR 17.5 · PCI 301 · B71 ✓	-81 dBm · SINR 16.9 · PCI 88 · B13 ✓	pass
1-14	-82 dBm · SINR 16.7 · PCI 212 · B2 ✓	-80 dBm · SINR 16 · PCI 301 · B71 ✓	-81 dBm · SINR 17.1 · PCI 88 · B13 ✓	pass

Square	AT&T, below 3.3 GHz	T-Mobile, below 3.3 GHz	Verizon, below 3.3 GHz	Cellular
1-15	-76 dBm · SINR 17.2 · PCI 212 · B2 ✓	-75 dBm · SINR 18.8 · PCI 301 · B71 ✓	-79 dBm · SINR 16.4 · PCI 88 · B13 ✓	pass
1-16	-65 dBm · SINR 23.4 · PCI 212 · B2 ✓	-63 dBm · SINR 27.2 · PCI 301 · B71 ✓	-64 dBm · SINR 24.5 · PCI 88 · B13 ✓	pass
1-17	-71 dBm · SINR 22.9 · PCI 212 · B2 ✓	-70 dBm · SINR 21.4 · PCI 301 · B71 ✓	-72 dBm · SINR 20.2 · PCI 88 · B13 ✓	pass
1-18	-81 dBm · SINR 18.8 · PCI 212 · B2 ✓	-77 dBm · SINR 17 · PCI 301 · B71 ✓	-78 dBm · SINR 19 · PCI 88 · B13 ✓	pass
1-19	-78 dBm · SINR 17.6 · PCI 212 · B2 ✓	-74 dBm · SINR 19.8 · PCI 301 · B71 ✓	-77 dBm · SINR 18.8 · PCI 88 · B13 ✓	pass
1-20	-78 dBm · SINR 18.1 · PCI 212 · B2 ✓	-76 dBm · SINR 18.8 · PCI 301 · B71 ✓	-77 dBm · SINR 18.5 · PCI 88 · B13 ✓	pass

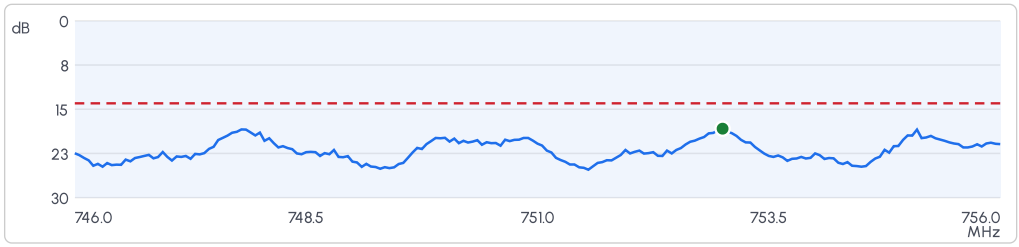
PIM and sweeps

Run because the owner or a carrier asked for them.

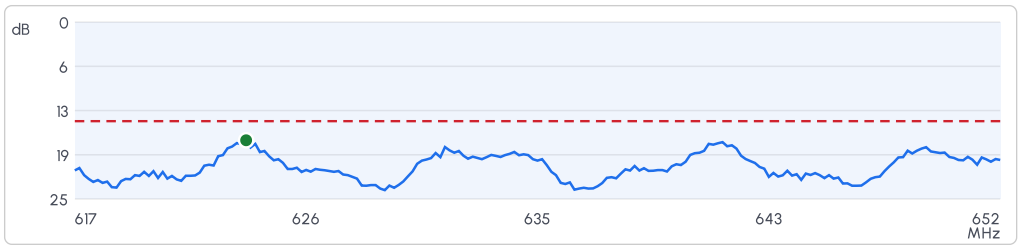
Sweep	Measurement	Instrument	Worst	Where on the trace	Limit and source	Result
Aisle trunk P1-P4, B2	Return loss	Sample site analyzer EX-4001 (example) · 2026-09-15	**17.4 dB RL**	1941 MHz	return loss ≥ 14 dB (VSWR ≤ 1.5:1): VSWR 1.5:1 (14 dB return loss) (The VSWR 1.5:1 rating on public-safety coax (e.g. the 2-hour rated coax in the parts research): common practice, not a code rule.)	pass
Aisle trunk P5-P8, B13	Return loss	Sample site analyzer EX-4001 (example) · 2026-09-15	**18.3 dB RL**	753.00 MHz	return loss ≥ 14 dB (VSWR ≤ 1.5:1): VSWR 1.5:1 (14 dB return loss) (The VSWR 1.5:1 rating on public-safety coax (e.g. the 2-hour rated coax in the parts research): common practice, not a code rule.)	pass
Aisle trunk P9-P12, B71	Return loss	Sample site analyzer EX-4001 (example) · 2026-09-15	**16.7 dB RL**	623.48 MHz	return loss ≥ 14 dB (VSWR ≤ 1.5:1): VSWR 1.5:1 (14 dB return loss) (The VSWR 1.5:1 rating on public-safety coax (e.g. the 2-hour rated coax in the parts research): common practice, not a code rule.)	pass



Aisle trunk P1-P4, B2 · Return loss · worst 17.4 dB (VSWR 1.31) at 1941 MHz · export:



Aisle trunk P5-P8, B13 · Return loss · worst 18.3 dB (VSWR 1.28) at 753.00 MHz · export:



Aisle trunk P9-P12, B71 · Return loss · worst 16.7 dB (VSWR 1.34) at 623.48 MHz · export:

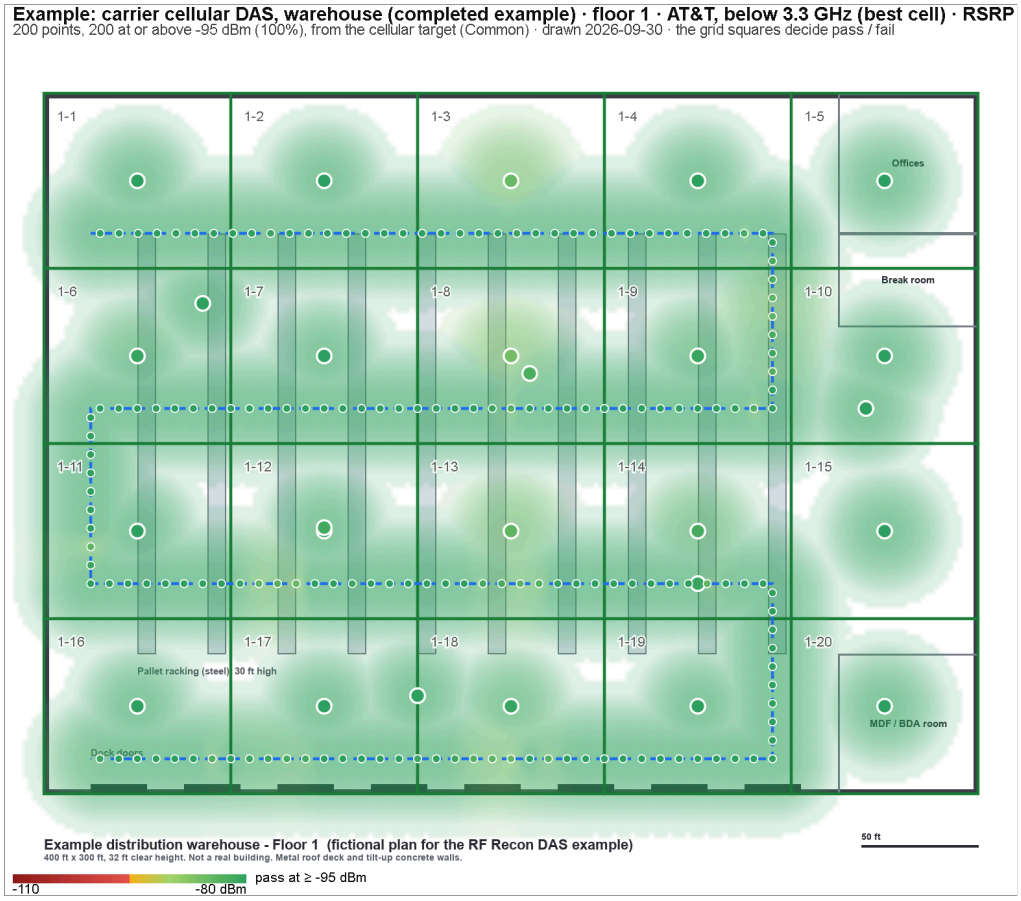
PIM runs

Where	Band	F1	F2	Tone dBm	Order	PIM dBm	dBc	Result (≤ -150 dBc)
Booster output, before the splitters	B2	**1935**	**1985**	**43**	IM3	**-110**	**-153.0**	pass
Aisle trunk, panels P1-P4	B2	**1935**	**1985**	**43**	IM3	**-109**	**-152.0**	pass
Aisle trunk, panels P5-P8	B13	**747**	**755**	**43**	IM3	**-111**	**-154.0**	pass
Aisle trunk, panels P9-P12	B71	**618**	**651**	**43**	IM3	**-108**	**-151.0**	pass

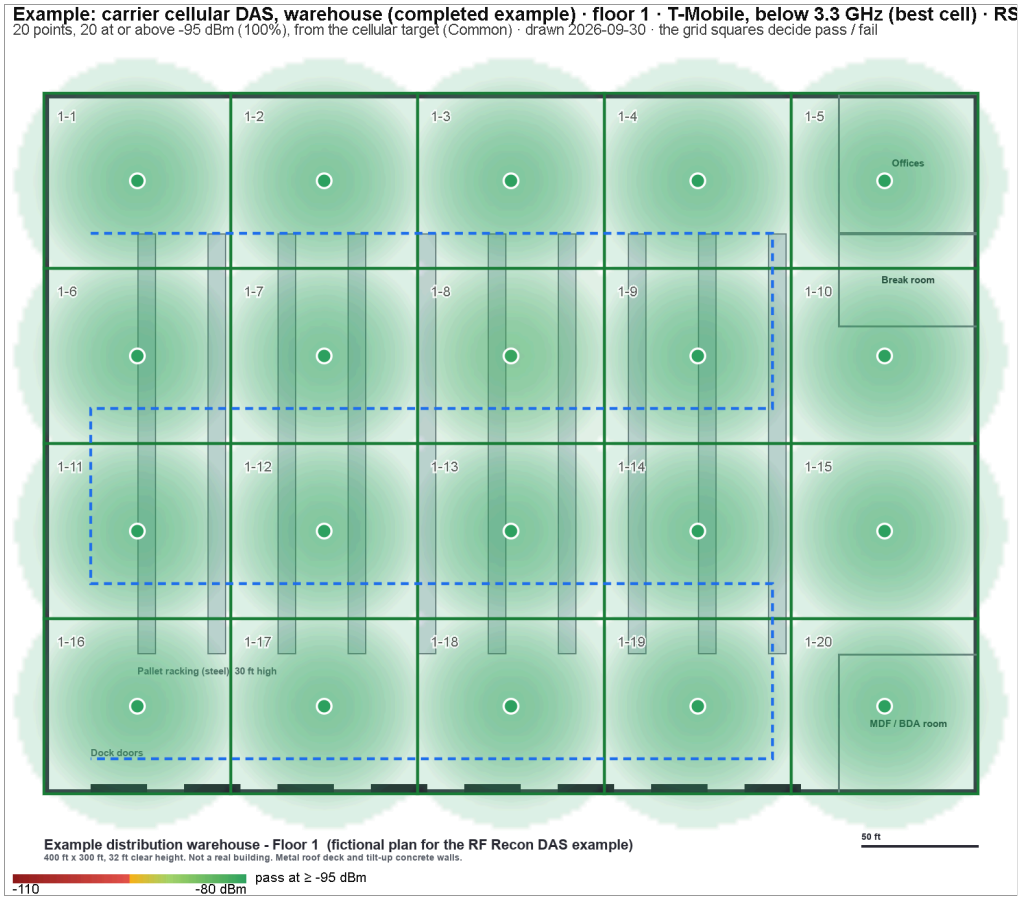
PIM limit -150 dBc: the job's own (PIM tab); −140 dBc at 2 × 43 dBm is the common system-level pass in the PIM research, not a code rule.

Walk test heat maps

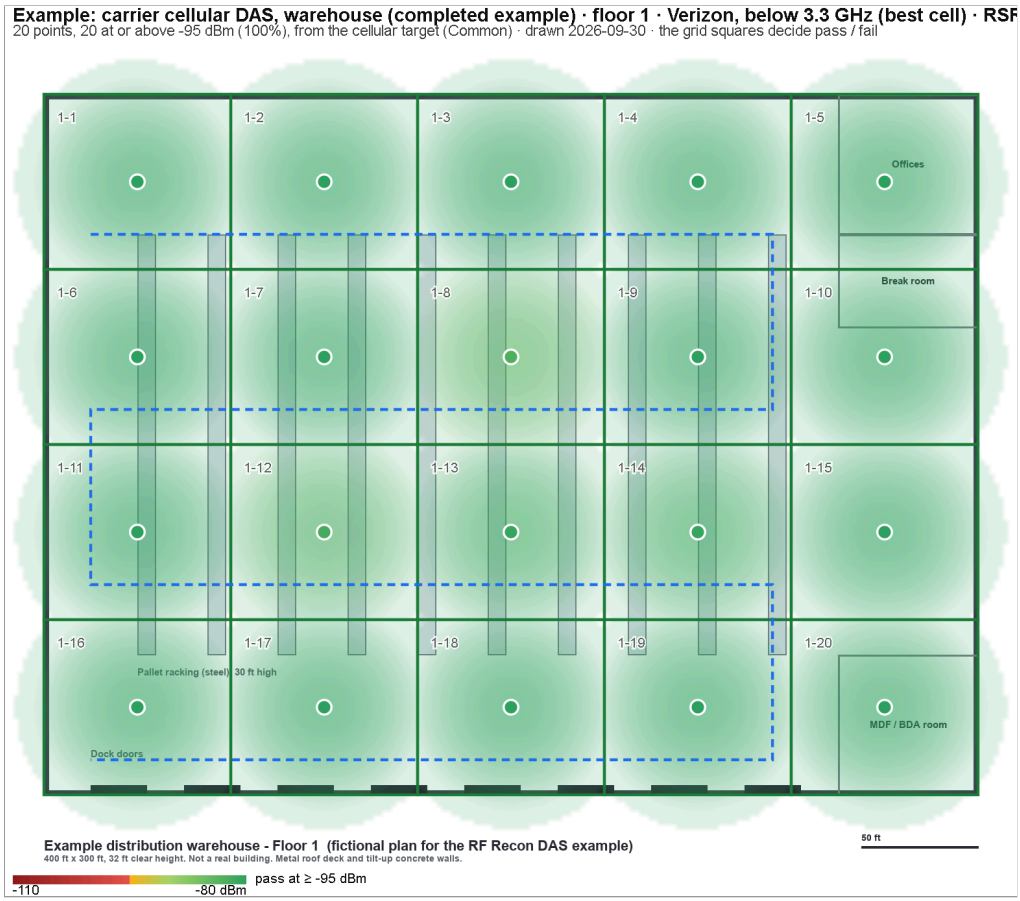
Readings walked on each floor's plan, coloured against the requirement. The heat map shows every reading, calibrated or not; the grid squares above decide pass / fail.



Floor 1 · AT&T, below 3.3 GHz (best cell) · RSRP dBm · 200 points, 200 at or above -95 dBm (100%), from the cellular target (Common) · drawn 2026-09-30 · the grid squares decide pass / fail



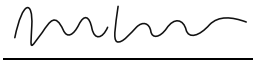
Floor 1 · T-Mobile, below 3.3 GHz (best cell) · RSRP dBm · 20 points, 20 at or above -95 dBm (100%), from the cellular target (Common) · drawn 2026-09-30 · the grid squares decide pass / fail



Floor 1 · Verizon, below 3.3 GHz (best cell) · RSRP dBm · 20 points, 20 at or above -95 dBm (100%), from the cellular target (Common) · drawn 2026-09-30 · the grid squares decide pass / fail

Tester's statement

I tested this building's commercial cellular coverage on 2026-09-22 to 2026-09-23 with the calibrated instruments listed in this report, against Owner spec: carrier coverage, Common preset. The results are the readings taken; none were changed.

Signature	Name, Sample RF Services (example)	Date
	Jordan Lee, RF technician, FCC GROL	2026-09-30

This product uses the FCC Data API but is not endorsed or certified by the FCC.

Sealed 2026-09-30 · verify code EX26-0930 · SHA-256 7b5ac602adea3683. Sealed copy of this report kept with the job; a changed job no longer matches it.